

Heavy-Tail First-Order Variance Correction

For the RSI log-odds coordinate under a finite $(3 + \delta)$ moment

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Abstract

This note extends the first law-dependent finite-horizon variance correction from exponentially light tails to symmetric innovation laws with a finite $(3 + \delta)$ absolute moment and a logarithmic lower-tail condition. The result is

$$\text{Var}(L_\alpha) = 2m_2\alpha + \frac{15m_2^2 + 2m_2 - 8m_3}{3}\alpha^2 + o(\alpha^2),$$

and therefore

$$g_F^*(n) = \frac{C_F^2}{\text{Var}(L_{1/n})} = n - K_F + o(1), \quad K_F = \frac{15m_2^2 + 2m_2 - 8m_3}{6m_2}.$$

For variance-normalized Student- t_ν innovations this applies for every $\nu > 3$ and gives an explicit K_ν . The proof uses a shrinking big-jump cutoff and termwise-truncated centered variables, so no fourth raw moment is required when $3 < p < 4$. The expansion is pointwise in the innovation law: its constants and rates may depend on δ and $\mathbb{E}R^{3+\delta}$, and no uniformity as $\delta \downarrow 0$ is claimed.

Verification scope

This is a conventional mathematical proof, not a theorem accepted by the Lean kernel. The accompanying Lean repository checks the exact finite-path endpoint identities, exponential-weight algebra, conditional log-odds transfer results, and a finite-variance central limit theorem for normalized *linear* Wilder rows. It does not currently prove the nonlinear variance expansion stated in this note.

The repository's pinned SymPy calculation independently reconstructs the formal moment and cumulant coefficients and their law-specific substitutions. It checks the finite coefficient algebra; the tail localization, indicator removal, and analytic remainder estimates are established by the argument below and are not machine-verified.

1 Setup

Let $\{\varepsilon_j\}_{j \geq 0}$ be i.i.d., symmetric, nondegenerate, and satisfy $\mathbb{P}(\varepsilon = 0) = 0$. Put

$$\mu_1 = \mathbb{E}|\varepsilon|, \quad R = \frac{|\varepsilon|}{\mu_1}, \quad Q = \text{sgn}(\varepsilon).$$

Symmetry and absence of an atom at zero imply that Q is Rademacher and independent of R . Hence $\mathbb{E}R = 1$. For $\alpha \in (0, 1)$ define geometric weights

$$w_j = \alpha(1 - \alpha)^j, \quad \sum_{j \geq 0} w_j = 1,$$

and

$$x = \sum_{j \geq 0} w_j Q_j R_j, \quad y = \sum_{j \geq 0} w_j (R_j - 1).$$

Then the signed directional balance is

$$F_\alpha = \frac{x}{1+y},$$

and the full RSI log-odds coordinate is

$$L_\alpha = 2 \operatorname{atanh}(F_\alpha).$$

The law of L_α is symmetric, hence $\mathbb{E}L_\alpha = 0$ whenever L_α is integrable.

Write

$$m_k = \mathbb{E}R^k, \quad s_r(\alpha) = \sum_{j \geq 0} w_j^r = \frac{\alpha^r}{1 - (1 - \alpha)^r}.$$

In particular,

$$s_2 = \frac{\alpha}{2} + \frac{\alpha^2}{4} + O(\alpha^3), \quad s_3 = \frac{\alpha^2}{3} + O(\alpha^3).$$

2 Main statement

Theorem 1 (First finite-horizon variance correction without exponential tails). *Assume that for some $\delta > 0$*

$$\mathbb{E}R^{3+\delta} < \infty.$$

Assume also that

$$\mathbb{E}[(-\log R)_+^q] < \infty$$

for some

$$q > \frac{2(2+\delta)}{\delta}.$$

Given such a q , choose $\gamma > 0$ so small that

$$0 < \gamma < \min \left\{ \frac{\delta}{3+\delta}, \frac{2+\delta-2q/(q-2)}{3+\delta} \right\}. \quad (1)$$

The second upper bound is positive precisely because of the strict inequality imposed on q . Then, as $\alpha \downarrow 0$,

$$\boxed{\operatorname{Var}(L_\alpha) = 2m_2\alpha + \frac{15m_2^2 + 2m_2 - 8m_3}{3}\alpha^2 + o(\alpha^2).} \quad (2)$$

Consequently, with

$$C_F^2 = 2m_2 = \frac{2\mathbb{E}\varepsilon^2}{(\mathbb{E}|\varepsilon|)^2},$$

the exact-variance normalizer satisfies

$$\boxed{g_F^*(n) := \frac{C_F^2}{\operatorname{Var}(L_{1/n})} = n - K_F + o(1),} \quad (3)$$

where

$$\boxed{K_F = \frac{15m_2^2 + 2m_2 - 8m_3}{6m_2}.} \quad (4)$$

Pointwise nature of the expansion. For each fixed innovation law satisfying the stated assumptions, one may fix an admissible $\delta > 0$, then choose q and γ as above. The remainder $o(\alpha^2)$ is therefore pointwise in the law. Neither the implied constants nor the rate are asserted to be uniform over a family for which $\delta \downarrow 0$ or $\mathbb{E}R^{3+\delta}$ diverges.

The proof is split into localization, complement control, and local moment algebra.

3 Weight and truncation estimates

Fix

$$p = 3 + \delta.$$

Choose γ according to (1) and define

$$\beta = p - 1 - \gamma p = 2 + \delta - \gamma(3 + \delta).$$

Then

$$\beta > \frac{2q}{q-2} > 2, \quad \beta \left(1 - \frac{2}{q}\right) > 2.$$

Choose q first and then a sufficiently small positive γ . Put

$$a_\alpha = \alpha^\gamma, \quad H_\alpha = \left\{ \sup_{j \geq 0} w_j R_j > a_\alpha \right\}.$$

Lemma 1 (Master tail estimates). *For $0 \leq r \leq p$,*

$$\sum_j w_j^r \mathbb{E} \left[R^r \mathbf{1}_{\{w_j R > a_\alpha\}} \right] \leq \mathbb{E} R^p a_\alpha^{r-p} s_p = O\left(\alpha^{p-1-\gamma(p-r)}\right). \quad (5)$$

For $r \geq p$,

$$\sum_j w_j^r \mathbb{E} \left[R^r \mathbf{1}_{\{w_j R \leq a_\alpha\}} \right] \leq \mathbb{E} R^p a_\alpha^{r-p} s_p = O\left(\alpha^{p-1+\gamma(r-p)}\right). \quad (6)$$

In particular,

$$\mathbb{P}(H_\alpha) = O(\alpha^\beta) = o(\alpha^2). \quad (7)$$

Moreover, the lost first, second, and third weighted moments caused by truncation are all $o(\alpha^2)$.

Proof. On $\{R > a/w_j\}$ and for $r \leq p$,

$$R^r \leq R^p (a/w_j)^{r-p}.$$

For $r \geq p$ the same inequality holds on $\{R \leq a/w_j\}$ with the inequality direction appropriate to $r - p \geq 0$. Summation gives (5)–(6). Setting $r = 0$ and using a union bound gives (7). For $r = 1, 2, 3$, the three exponents in (5) are, respectively,

$$2 + \delta - \gamma(2 + \delta), \quad 2 + \delta - \gamma(1 + \delta), \quad 2 + \delta - \gamma\delta.$$

Each is strictly larger than 2 under $\gamma < \delta/(3 + \delta)$. Hence the corresponding lost weighted moments are $o(\alpha^2)$. $\square$

4 Localization without an exponential moment

Define termwise-truncated magnitudes

$$\tilde{R}_j = R_j \mathbf{1}_{\{w_j R_j \leq a_\alpha\}},$$

and centered truncated variables

$$A_j = Q_j \tilde{R}_j, \quad B_j = \tilde{R}_j - \mathbb{E} \tilde{R}_j.$$

Set

$$\tilde{x} = \sum_j w_j A_j, \quad z = \sum_j w_j B_j, \quad b_\alpha = \sum_j w_j (\mathbb{E} \tilde{R}_j - 1).$$

By symmetry $\mathbb{E}A_j = 0$, while Lemma 1 with $r = 1$ gives

$$|b_\alpha| \leq \sum_j w_j \mathbb{E}[R_j \mathbf{1}_{\{w_j R_j > a_\alpha\}}] = o(\alpha^2). \quad (8)$$

On H_α^c we have exactly

$$x = \tilde{x}, \quad y = z + b_\alpha. \quad (9)$$

The event H_α^c bounds each weighted summand $w_j R_j$ by a_α ; it does not imply $|x| \leq a_\alpha$ or $|y| \leq a_\alpha$. The infinite sums are controlled below through variance and moment bounds.

Lemma 2 (Uniform moments of the truncated vector). *For every fixed $r \geq 2$,*

$$\sup_{0 < \alpha < \alpha_0} (\mathbb{E}|\tilde{x}|^r + \mathbb{E}|z|^r) < \infty.$$

More precisely, Rosenthal's inequality bounds each moment by a constant multiple of a variance term of order $O(\alpha^{r/2})$ plus a diagonal term. For $r \leq p$ the diagonal term is $O(s_r) = O(\alpha^{r-1})$; for $r > p$ Lemma 1 gives

$$\sum_j w_j^r \mathbb{E} \tilde{R}_j^r = O\left(\alpha^{p-1+\gamma(r-p)}\right).$$

The same conclusion holds for B_j after centering.

Proof. Apply Rosenthal's inequality first to finite partial sums. The sum of variances is $O(\alpha)$ because $\sum_j w_j^2 = O(\alpha)$ and the truncated second moments are uniformly bounded. The stated diagonal bounds follow from the finite p -moment when $r \leq p$ and from (6) when $r > p$. Letting the number of summands tend to infinity gives the result by L^r convergence. $\square$

The centered weighted summands $w_j A_j$ have absolute size at most a_α , and $w_j B_j$ have absolute size at most $a_\alpha + O(\alpha) = O(a_\alpha)$. Their total variances are $O(\alpha)$.

Lemma 3 (Shrinking-cutoff concentration). *For each fixed sufficiently small $\eta > 0$ there are constants $c, C > 0$ such that*

$$\mathbb{P}((|x| > \eta \text{ or } |y| > \eta) \cap H_\alpha^c) \leq C \exp(-c\alpha^{-\gamma}) \quad (10)$$

for all sufficiently small α . Consequently, if

$$G_\alpha = H_\alpha^c \cap \{|x| \leq \eta, |y| \leq \eta\},$$

then

$$\mathbb{P}(G_\alpha^c) = O(\alpha^\beta) + O(e^{-c\alpha^{-\gamma}}). \quad (11)$$

Proof. Apply Bernstein's inequality to finite partial sums of the independent centered variables $w_j A_j$ and $w_j B_j$. The variance proxy is $O(\alpha)$ and the maximal summand is $O(a_\alpha) = O(\alpha^\gamma)$. For a fixed deviation $u > 0$,

$$2 \exp \left[-c \min \left(\frac{u^2}{\alpha}, \frac{u}{\alpha^\gamma} \right) \right] \leq 2e^{-c'\alpha^{-\gamma}},$$

since $0 < \gamma < 1$. Pass to the infinite sums by L^2 convergence of the partial sums. By (8), $y = z + b_\alpha$ on H_α^c with $b_\alpha = o(1)$, so the deterministic bias is absorbed into u for small α . Add (7). $\square$

Why the cutoff must shrink. A fixed cutoff $a_\alpha \equiv a > 0$ would only make the Bernstein bounded-summand term $\exp(-c/a)$, which does not vanish with α . The shrinking cutoff $a_\alpha = \alpha^\gamma$ is therefore essential in the polynomial-tail proof.

5 The complement contributes below order α^2

Let T_+ and T_- be the first indices with $Q_j = +1$ and $Q_j = -1$, respectively. They are geometric with success probability $1/2$ and are independent of the magnitudes. Since

$$U \geq \mu_1 w_{T_+} R_{T_+}, \quad D \geq \mu_1 w_{T_-} R_{T_-},$$

we obtain

$$(-\log U)_+ \leq C + \log(1/\alpha) + T_+[-\log(1-\alpha)] + (-\log R_{T_+})_+,$$

and similarly for D . Positive logarithms are uniformly controlled by the finite p -moment, because $U, D \leq \mu_1 \sum_j w_j R_j$ and Jensen's inequality gives a uniform positive moment of the weighted average.

Lemma 4 (Logarithmic moments and bad-set negligibility). *Under the assumptions of Theorem 1,*

$$\mathbb{E}|L_\alpha|^q = O(1 + \log^q(1/\alpha)). \quad (12)$$

Moreover,

$$\mathbb{E}[L_\alpha^2 \mathbf{1}_{G_\alpha^c}] = o(\alpha^2). \quad (13)$$

Proof. The first-success bounds above, the finite q -moment of $(-\log R)_+$, and the geometric moments of $T_\pm$ yield (12). By Hölder and (11),

$$\mathbb{E}[L_\alpha^2 \mathbf{1}_{G_\alpha^c}] \leq (\mathbb{E}|L_\alpha|^q)^{2/q} \mathbb{P}(G_\alpha^c)^{1-2/q}.$$

The polynomial contribution is $O(\log^2(1/\alpha)\alpha^{\beta(1-2/q)}) = o(\alpha^2)$ by construction. The stretched-exponential contribution is smaller than every power of α . $\square$

6 Local Taylor algebra and moment bookkeeping

Choose $\eta < 1/3$. On G_α ,

$$\left| \frac{x}{1+y} \right| \leq \frac{\eta}{1-\eta} < \frac{1}{2},$$

so the Taylor series is uniformly regular. Direct expansion gives

$$\frac{L_\alpha^2}{4} = x^2 - 2x^2y + 3x^2y^2 + \frac{2}{3}x^4 + \mathcal{R}_5, \quad (14)$$

where on G_α

$$|\mathcal{R}_5| \leq C_\eta(|x| + |y|)^5.$$

Why total degree four is sufficient. On the fixed neighborhood defining G_α , the map $2 \operatorname{atanh}(x/(1+y))$ is analytic and the displayed polynomial contains every term in its square through total degree four. Terms of total degree at least five are therefore absorbed into $\mathcal{R}_5$. Consequently, no such term can affect the coefficient of α^2 once

$$\mathbb{E}[(|x| + |y|)^5 \mathbf{1}_{G_\alpha}] = o(\alpha^2)$$

has been established. Lemma 5 proves precisely this analytic remainder condition; the symbolic calculation checks only the resulting finite polynomial algebra.

Lemma 5 (Local moments through total degree four). *Under the assumptions of Theorem 1,*

$$\mathbb{E}[x^2 \mathbf{1}_{G_\alpha}] = m_2 s_2 + o(\alpha^2), \quad (15)$$

$$\mathbb{E}[x^2 y \mathbf{1}_{G_\alpha}] = (m_3 - m_2) s_3 + o(\alpha^2), \quad (16)$$

$$\mathbb{E}[x^4 \mathbf{1}_{G_\alpha}] = 3m_2^2 s_2^2 + o(\alpha^2), \quad (17)$$

$$\mathbb{E}[x^2 y^2 \mathbf{1}_{G_\alpha}] = m_2(m_2 - 1) s_2^2 + o(\alpha^2), \quad (18)$$

$$\mathbb{E}[(|x| + |y|)^5 \mathbf{1}_{G_\alpha}] = o(\alpha^2). \quad (19)$$

Proof. We work with the centered termwise-truncated vector $(\tilde{x}, z)$ from Section 4. This does not require an untruncated fourth moment when $3 < p < 4$.

Removing the indicators. Let P be any fixed polynomial in two variables of total degree at most five. Lemma 2 gives uniformly bounded moments of $P(\tilde{x}, z)$ of arbitrarily high fixed order. Since $\beta > 2$, choose $s > 1$ so large that $\beta(1 - 1/s) > 2$. Hölder therefore gives

$$\mathbb{E}[|P(\tilde{x}, z)|\mathbf{1}_{H_\alpha}] \leq (\mathbb{E}|P(\tilde{x}, z)|^s)^{1/s} \mathbb{P}(H_\alpha)^{1-1/s} = o(\alpha^2).$$

On H_α^c , identities (9) hold. The additional localization failure $\{|x| > \eta \text{ or } |y| > \eta\} \cap H_\alpha^c$ has stretched-exponentially small probability by Lemma 3, and another Hölder bound using Lemma 2 makes its contribution $o(\alpha^M)$ for every fixed M . Thus, for the polynomials below, expectations localized to G_α may be replaced, up to $o(\alpha^2)$, by full expectations of the corresponding expressions in $(\tilde{x}, z + b_\alpha)$.

Second and third degree. Put

$$V_A = \sum_j w_j^2 \mathbb{E} \tilde{R}_j^2, \quad V_B = \sum_j w_j^2 \text{Var}(\tilde{R}_j).$$

The $r = 1, 2$ tail bounds in Lemma 1 imply

$$V_A = m_2 s_2 + o(\alpha^2), \quad V_B = (m_2 - 1) s_2 + o(\alpha^2). \quad (20)$$

Hence $\mathbb{E} \tilde{x}^2 = V_A$. Because the signs are independent and centered, every term in $\mathbb{E}(\tilde{x}^2 z)$ vanishes except the same-index triple, so

$$\mathbb{E}(\tilde{x}^2 z) = \sum_j w_j^3 \mathbb{E} \left[\tilde{R}_j^2 (\tilde{R}_j - \mathbb{E} \tilde{R}_j) \right] = (m_3 - m_2) s_3 + o(\alpha^2),$$

where the error follows from the $r = 1, 2, 3$ tail bounds. Since $b_\alpha = o(\alpha^2)$ and $V_A = O(\alpha)$,

$$\mathbb{E}[\tilde{x}^2(z + b_\alpha)] = \mathbb{E}(\tilde{x}^2 z) + o(\alpha^2).$$

This proves (15)–(16).

Fourth moment of x . For independent centered variables $w_j A_j$,

$$\mathbb{E} \tilde{x}^4 = 3V_A^2 + D_4,$$

where

$$D_4 = \sum_j w_j^4 \left(\mathbb{E} \tilde{R}_j^4 - 3(\mathbb{E} \tilde{R}_j^2)^2 \right).$$

If $p < 4$ (equivalently, $0 < \delta < 1$), (6) gives

$$\sum_j w_j^4 \mathbb{E} \tilde{R}_j^4 = O\left(\alpha^{p-1+\gamma(4-p)}\right).$$

Writing $p = 3 + \delta$, the exponent is

$$p - 1 + \gamma(4 - p) = 2 + \delta + \gamma(1 - \delta) > 2$$

because $1 - \delta > 0$, so this term is $o(\alpha^2)$. More explicitly,

$$\frac{1}{\alpha^2} \sum_j w_j^4 \mathbb{E} \tilde{R}_j^4 = O\left(\alpha^{\delta+\gamma(1-\delta)}\right) \rightarrow 0.$$

This estimate is not uniform as $\delta \downarrow 0$: its constant contains $\mathbb{E} R^{3+\delta}$ and its decay exponent can become arbitrarily small. If $p \geq 4$, the finite fourth moment gives $O(s_4) = O(\alpha^3)$. Also $\sum_j w_j^4 (\mathbb{E} \tilde{R}_j^2)^2 = O(s_4)$. Thus $D_4 = o(\alpha^2)$ and, by (20),

$$\mathbb{E} \tilde{x}^4 = 3m_2^2 s_2^2 + o(\alpha^2).$$

This proves (17). In particular, no untruncated fourth moment is invoked when $3 < p < 4$.

The mixed fourth moment. Because $\mathbb{E}(A_j B_j) = 0$ by the independent Rademacher sign, the pair-partition formula gives

$$\mathbb{E}(\tilde{x}^2 z^2) = V_A V_B + D_{22},$$

where D_{22} is a same-index fourth-degree diagonal. Its absolute value is bounded by a constant multiple of

$$\sum_j w_j^4 \left(\mathbb{E} \tilde{R}_j^4 + \mathbb{E} \tilde{R}_j^2 + 1 \right) = o(\alpha^2)$$

by the same diagonal estimate used for D_4 . Thus $D_{22} = o(\alpha^2)$ for each fixed law satisfying the assumptions, with the same non-uniformity near the third-moment boundary. Hence

$$\mathbb{E}(\tilde{x}^2 z^2) = m_2(m_2 - 1)s_2^2 + o(\alpha^2).$$

The deterministic truncation bias is harmless:

$$\mathbb{E}[\tilde{x}^2(z + b_\alpha)^2] = \mathbb{E}(\tilde{x}^2 z^2) + 2b_\alpha \mathbb{E}(\tilde{x}^2 z) + b_\alpha^2 V_A,$$

and the last two terms are $o(\alpha^2)$ by (8). After centering, the singleton- y configurations vanish; their residual contribution is contained in $b_\alpha = o(\alpha^2)$. Equation (18) follows.

Fifth-degree remainder. Rosenthal's inequality applied to the centered truncated sums gives

$$\mathbb{E}|\tilde{x}|^5 + \mathbb{E}|z|^5 \leq C \left[O(\alpha^{5/2}) + \sum_j w_j^5 \mathbb{E} \tilde{R}_j^5 + O(s_5) \right].$$

If $p < 5$ (equivalently, $0 < \delta < 2$), Lemma 1 bounds the diagonal term by

$$O(\alpha^{p-1+\gamma(5-p)}) = O(\alpha^{2+\delta+\gamma(2-\delta)}) = o(\alpha^2),$$

and if $p \geq 5$ it is $O(s_5) = O(\alpha^4)$. Together with $b_\alpha = o(\alpha^2)$ this yields

$$\mathbb{E}(|\tilde{x}| + |z + b_\alpha|)^5 = o(\alpha^2).$$

The indicator-removal argument at the start of the proof then gives (19). □

7 Proof of Theorem 1

By Lemma 4, the complement of G_α contributes $o(\alpha^2)$ to $\mathbb{E}L_\alpha^2$. On G_α , use (14), Lemma 5, and (19):

$$\frac{1}{4}\mathbb{E}L_\alpha^2 = m_2 s_2 - 2(m_3 - m_2)s_3 + 3m_2(m_2 - 1)s_2^2 + 2m_2^2 s_2^2 + o(\alpha^2).$$

Since

$$s_2 = \frac{\alpha}{2} + \frac{\alpha^2}{4} + O(\alpha^3), \quad s_3 = \frac{\alpha^2}{3} + O(\alpha^3), \quad s_2^2 = \frac{\alpha^2}{4} + O(\alpha^3),$$

we obtain

$$\frac{1}{4}\mathbb{E}L_\alpha^2 = \frac{m_2}{2}\alpha + \frac{15m_2^2 + 2m_2 - 8m_3}{12}\alpha^2 + o(\alpha^2).$$

Symmetry gives $\mathbb{E}L_\alpha = 0$, so (2) follows. Finally,

$$\text{Var}(L_{1/n}) = \frac{C_F^2}{n} \left(1 + \frac{K_F}{n} + o(n^{-1}) \right),$$

and inversion gives (3).

8 Symbolic coefficient check

The accompanying script `supplement/moment_expansion_audit.py` reconstructs the relevant formal moment and cumulant expressions with exact SymPy arithmetic. In particular, it independently reproduces

$$\frac{15m_2^2 + 2m_2 - 8m_3}{3}$$

as the coefficient of α^2 in (2), together with the Laplace, Rademacher, Gaussian, and Student-law substitutions used elsewhere in the project. The script is run by continuous integration.

This reproduces the finite coefficient calculation but is not a second proof of Theorem 1. The theorem additionally depends on the localization, logarithmic-moment, indicator-removal, and Taylor-remainder arguments in the preceding sections.

9 Student- t corollary

Corollary 1 (Variance-normalized Student- t_ν , $\nu > 3$). *Let ε be a centered Student- t_ν random variable, rescaled to variance one, with $\nu > 3$. Then all assumptions of Theorem 1 hold (choose any $p \in (3, \nu)$). Moreover*

$$C_\nu = \sqrt{\frac{2\pi}{\nu-2}} \frac{\Gamma(\nu/2)}{\Gamma((\nu-1)/2)} \quad (21)$$

and

$$K_\nu = \frac{5}{4}C_\nu^2 - \frac{7}{3} - \frac{8}{3(\nu-3)}. \quad (22)$$

Hence, for each fixed $\nu > 3$,

$$g_\nu^*(n) = n - K_\nu + o(1).$$

The remainder is pointwise in ν ; no uniformity as $\nu \downarrow 3$ is asserted.

Proof. For raw Student- t_ν ,

$$\mathbb{E}|T|^r = \nu^{r/2} \frac{\Gamma((r+1)/2)\Gamma((\nu-r)/2)}{\sqrt{\pi}\Gamma(\nu/2)}, \quad r < \nu.$$

Scale cancels from $R = |\varepsilon|/\mathbb{E}|\varepsilon|$. Formula (21) follows from $C_\nu^2 = 2m_2$. Also

$$\frac{m_3}{m_2} = \frac{\mathbb{E}|T|^3}{\mathbb{E}|T|\mathbb{E}T^2} = \frac{2(\nu-2)}{\nu-3}.$$

Substitution into (4) gives (22). Student- t has finite logarithmic moments near zero and at infinity, so the logarithmic assumption is automatic. For a given $\nu > 3$, one may choose any fixed δ with $0 < \delta < \nu - 3$. The constants necessarily deteriorate as $\nu \downarrow 3$, because the available δ shrinks and $\mathbb{E}|T|^{3+\delta}$ diverges as $3 + \delta \uparrow \nu$. $\square$

Selected values are

ν	C_ν	K_ν
4	2	0
4.5	1.954366	0.663323
5	1.923825	0.959710
5.1013...	1.918833	1
6	1.885618	11/9
8	1.847521	7/5
∞ (Gaussian)	$\sqrt{\pi}$	1.593657...

The value $\nu = 4$ is a genuine cancellation point: $K_4 = 0$, so the first constant correction vanishes and

$$g_4^*(n) = n + o(1).$$

This does not mean that the finite- n normalization is exactly n ; it means only that the constant-shift term is zero. At the opposite edge, K_ν is not bounded as $\nu \downarrow 3^+$ because of the pole $-8/[3(\nu-3)]$. This divergence is consistent with, and warns against, any uniform interpretation of the $o(1)$ remainder across the full family $\nu > 3$.

The familiar proxy $\sqrt{n-1}$ is therefore not a Student- t law in general. Within this family it is first-order variance-correct only at a root of $K_\nu = 1$; numerically,

$$\nu_* \approx 5.1013.$$

Thus the appearance of $n-1$ corresponds to a particular tail thickness, not to a universal geometric law.

10 Scope and limitations

This note does not address the boundary $\nu = 3$ or the regime $2 < \nu < 3$. The shrinking-cutoff proof requires a moment strictly above order three.

The note also does not derive higher-order coefficients under polynomial tails. It establishes only the constant shift K_F , generated by the α^2 term in the variance. Corrections of order $1/n$ and $1/n^2$ would require control of progressively higher diagonal moments and their threshold cases.

11 Conclusion

For every fixed Student- t_ν law with $\nu > 3$, the first-order variance-normalizing length has the form

$$g_\nu^*(n) = n - K_\nu + o(1),$$

with K_ν given by (22). This extends the constant-shift calculation beyond exponential tails, but the remainder is pointwise in ν and the result does not extend uniformly to the third-moment boundary.